

AP Cyber- security: The Complete Exam Guide

A new AP Career Kickstart course: the 5 units, the risk-and-detection skills, the evidence-based free-response question, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**2:10**

hours total
80 + 50 minutes

60+1

multiple-choice +
free-response question

70%

of your score is
multiple choice

New

AP Career Kickstart
course

INSIDE THIS GUIDE

01	The exam at a glance	2	06	10 traps that cost points	7
02	Section I: multiple choice	3	07	Your study roadmap to May	8
03	The units, in plain English	4	08	10 practice questions	9
04	Section II: free response	5	09	Answer key & explanations	11
05	Anatomy of a full-credit answer	6	10	Scores & exam-day checklist	12

01 THE EXAM AT A GLANCE

Two sections, one score

AP Cybersecurity is a new AP Career Kickstart course. Your AP score (1–5) combines a multiple-choice section with one in-depth analysis question built on real-world security data.

70%

Section I · Multiple choice · 60 questions

30%

Section II · Free response · 1 question

Exam structure

Section	Weight & timing	What you do
I · Multiple choice	70% of score 1 hr 20 min 60 questions	Individual questions and sets of 2–4 questions. Apply security concepts to real situations: identify vulnerabilities, recommend controls, and detect attacks.
II · Free response	30% of score 50 minutes 1 question	An analysis prompt built on multiple sources such as security policies, firewall configurations, system logs and access permissions. Cite evidence and explain your reasoning.

FULLY DIGITAL

Everything happens in Bluebook

You answer every question, including the written responses, in College Board's **Bluebook** testing app. Responses are submitted automatically at the end. Practice typing timed answers before May.

ETHICAL HACKING MINDSET

Think like an adversary, act ethically

The course teaches **adversarial thinking**: understanding how attackers work so you can defend systems. Always practice only on systems you are authorized to use.

How to use this guide

Just starting?

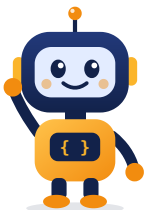
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 13 minutes, then use the final-four-weeks plan.



Pigo's tip: Practice **reading logs and configurations**. The free-response question rewards spotting evidence in technical data and explaining what it means.

SECTION I

02 The multiple-choice section

60 questions in 80 minutes gives you 1 minute 20 seconds per question. Many questions describe a scenario and ask for the best control or the most likely attack.

60

scenario-based questions

1:20

minutes per question on average

70%

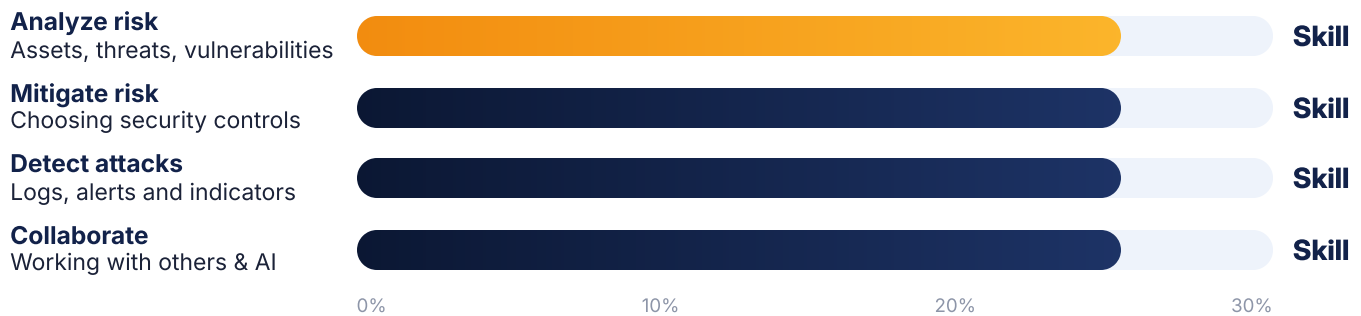
of your total AP score

0

points lost for a wrong answer, so never leave one blank

◆ The four skills tested

College Board organizes the course around four skills applied across all units. Unit weights are not yet published.

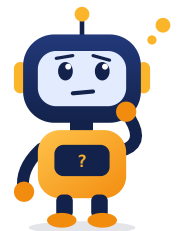


What this means for your study time: Every unit is tested through the same skills: analyze risk, mitigate it, and detect attacks. Practice applying each skill to physical spaces, networks, devices, and applications and data.

◆ Strategies that work on this section

- ◆ **Identify the asset** being protected first, then the threat and the vulnerability.
- ◆ **Use the CIA triad:** is confidentiality, integrity or availability at risk?
- ◆ **Pick the most direct control:** the one that addresses the specific vulnerability described.
- ◆ **Read logs line by line:** timestamps, IP addresses, users and repeated failures are clues.
- ◆ **Think defense in depth:** layered controls beat a single control.
- ◆ **Never leave a blank.** There is no penalty for guessing.

Ask: what is the **asset**, what is the **threat**, and what is the **weakness**? The answer usually follows.



03 WHAT THE COURSE COVERS

The 5 units, in plain English

The course moves from people and buildings to networks, devices and data.

UNIT

1

Introduction to Security

People, passwords and adversaries.

Be ready to: explain the CIA triad; strengthen authentication (strong passwords, multifactor); recognize social engineering such as phishing; describe types of cyber adversaries.

UNIT

2

Securing Spaces

Locks, cameras and access.

Be ready to: identify physical vulnerabilities and threats; recommend controls such as badges, locks and cameras; detect physical breaches with adversarial thinking.

UNIT

3

Securing Networks

Firewalls, segmentation and traffic.

Be ready to: explain network segmentation; configure and evaluate firewall rules; analyze network logs for indicators of compromise.

UNIT

4

Securing Devices

Computers, phones and IoT.

Be ready to: harden devices with updates, least privilege and anti-malware; analyze device logs for signs of compromise.

UNIT

5

Securing Applications & Data

Access control and encryption.

Be ready to: apply access controls and permissions; use encryption and hashing to protect data; detect attacks on applications and data such as injection.

New course: because AP Cybersecurity is new, national score data and released questions are still limited. Check College Board for the latest course updates each year.

SECTION II

04

The free-response question

50 minutes for one in-depth question. You analyze several technical sources and explain a security situation with evidence.

Analyze sources

Identify risk

Recommend controls

Detect & explain

One question worth 30% of your AP score

Src

Read every source

Policies, configs, logs

You may see a security policy, firewall rules, system logs or a permissions table. Each one holds evidence.

Annotate: mark anything that breaks the policy or looks unusual.

Risk

Identify the risk

Asset, threat, vulnerability

Explain what could go wrong, which asset is at risk, and which part of the CIA triad is affected.

Name the specific line or setting that creates the vulnerability.

Fix

Recommend controls

Mitigation

Recommend specific controls and explain how each reduces the risk you identified.

"Enable MFA for admin accounts" beats "improve security".

Det

Detect and explain

Indicators of compromise

Use log evidence to explain whether an attack occurred, what kind, and how you know.

Quote timestamps, IPs or usernames as evidence.

◆ A time plan for the 50 minutes

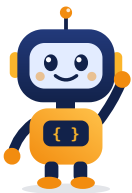
Read sources
10 min

Plan
5 min

Write
30 min

Check
5 min

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Every claim needs **evidence from a source**: a log line, a rule number or a permission setting.

05 FREE-RESPONSE PLAYBOOK

Anatomy of a strong analysis

Here is a short log-analysis practice prompt with a strong response, annotated to show where points come from.

PRACTICE TASK · WRITTEN BY ACT SAT GO

Practice prompt: A school's policy says accounts lock after 5 failed logins. A log shows 240 failed logins for the user "admin" from IP 203.0.113.7 between 02:10 and 02:14, followed by one successful login from the same IP at 02:15. The account was never locked. Explain what likely happened, identify the vulnerability, and recommend two controls.

Sample response

- ① The log shows a likely **brute-force attack**: 240 failed logins in four minutes from one IP, then a successful login at 02:15.
- ② The account was never locked, even though policy requires lockout after 5 failures, so the **lockout control was misconfigured or disabled**.
- ③ This threatens **confidentiality and integrity**, because an attacker may now control an administrator account.
- ④ **Controls:** (1) enforce account lockout after 5 failures, stopping repeated guessing; (2) require multifactor authentication for admin accounts, so a guessed password alone isn't enough.

- ① **Name the attack** and cite the log evidence.
- ② **Identify the vulnerability** by comparing log and policy.
- ③ **Name the impact** using the CIA triad.
- ④ **Recommend specific controls** and explain how each helps.

◆ Five habits for the free-response question

- 1**
Read every source before writing.
- 2**
Quote log lines, rules or settings as evidence.
- 3**
Use the CIA triad to describe impact.
- 4**
Recommend specific, named controls.
- 5**
Explain how each control reduces the risk.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Encryption fixes **everything!** (It doesn't.)



01 Vague controls

"Improve security" earns nothing. Name the control and how it works.

02 Encryption vs hashing

Encryption is reversible with a key; hashing is one-way and used for passwords and integrity checks.

03 Authentication vs authorization

Authentication proves who you are; authorization decides what you may access.

04 Ignoring the policy

Compare logs and configurations against the stated policy to find violations.

05 Firewall rule order

Firewalls usually apply the first matching rule, so a broad "allow" above a "deny" can override it.

06 Single point of defense

Defense in depth layers controls so one failure doesn't expose everything.

07 Least privilege forgotten

Users should have only the access they need. Excess permissions are a common vulnerability.

08 Missing the human factor

Many attacks begin with social engineering. Training is a control too.

09 No evidence

In the free-response question, every claim needs evidence from a source.

10 Unauthorized practice

Never test attacks on systems you don't own or have permission to use.

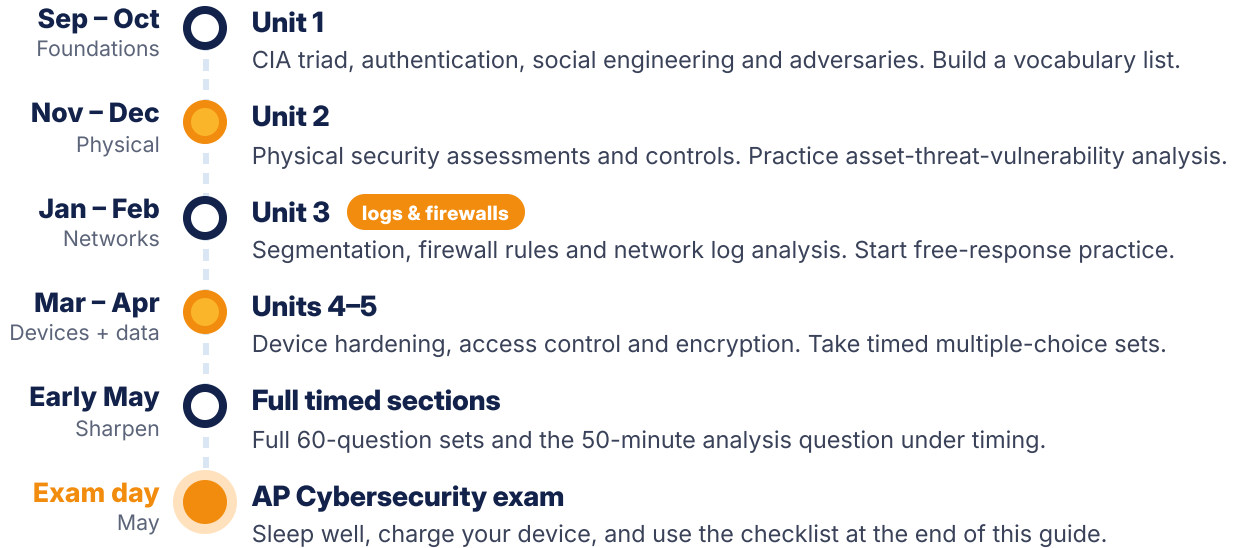
Quick self-check: for a stolen laptop containing student records, can you name the asset, the threat, the vulnerability, the CIA impact and two controls? (e.g., full-disk encryption and remote wipe.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Practice reading logs and configurations every week.



• The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 30 multiple-choice questions plus one analysis question	List your weakest unit
3 weeks out	Daily log and firewall-rule reading	Find the violation every time
2 weeks out	Full practice exam under timing	Finish both sections
Final week	Light review: vocabulary, controls, traps list	Rest the night before

08 TEST YOURSELF

10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **13 minutes** for all ten, then check page 11.

1 UNIT 1 • CIA TRIAD

A ransomware attack encrypts a hospital's patient records so staff can't open them. Which part of the CIA triad is most directly affected?

- ☐ A Availability
- ☐ B Confidentiality
- ☐ C Integrity
- ☐ D Authentication

2 UNIT 1 • SOCIAL ENGINEERING

An email that appears to come from the school IT team asks students to "verify" their password through a link. This is:

- ☐ A a firewall breach
- ☐ B phishing
- ☐ C a denial-of-service attack
- ☐ D encryption

3 UNIT 1 • AUTHENTICATION

Which login method is strongest?

- ☐ A A 6-character password
- ☐ B A password written on a sticky note
- ☐ C The same password on every site
- ☐ D A long passphrase plus a one-time code from an authenticator app

4 UNIT 2 • PHYSICAL SECURITY

An employee holds the door open for someone without a badge. This security failure is called:

- ☐ A hashing
- ☐ B spoofing
- ☐ C tailgating
- ☐ D patching

5 UNIT 3 • FIREWALLS

A firewall applies the first rule that matches.

Rule	Action	Source	Port
1	ALLOW	Any	Any
2	DENY	Any	23 (Telnet)
3	ALLOW	Any	443 (HTTPS)

What is the problem with this rule set?

- ☐ A HTTPS is blocked
- ☐ B Telnet is allowed only from trusted IPs
- ☐ C There is no problem
- ☐ D Rule 1 allows all traffic, so rule 2 never blocks Telnet

6 UNIT 3 • SEGMENTATION

A school places its security cameras on a separate network from student laptops. The main benefit is that:

- ☐ A the cameras record in higher quality
- ☐ B a compromised laptop can't easily reach the cameras
- ☐ C the internet is faster
- ☐ D passwords are no longer needed

08 TEST YOURSELF (continued)

7 UNIT 4 · DEVICES

Which control most directly protects data on a lost laptop?

- A A screen saver
- B A larger hard drive
- C Full-disk encryption
- D Faster Wi-Fi

8 UNIT 5 · HASHING

Why do secure systems store password hashes instead of passwords?

- A If the database is stolen, attackers don't directly see the passwords
- B Hashes are easier to read
- C Hashes can be reversed with a key
- D Hashes make logins faster

9 UNIT 5 · ACCESS CONTROL

A student-records clerk can also edit payroll data. Which principle is violated?

- A Defense in depth
- B Availability
- C Non-repudiation
- D Least privilege

10 DETECTION

A log shows 300 failed logins for one account in two minutes from the same IP. This most likely indicates:

- A a brute-force attack
- B a software update
- C a backup running
- D normal activity



CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	A · Availability	Encrypting the data so staff can't access it attacks availability. (Data may also be stolen, but the main effect is lost access.)
2	B · phishing	A fake message tricking users into revealing credentials is phishing, a form of social engineering.
3	D	A long passphrase plus multifactor authentication combines something you know with something you have.
4	C · tailgating	Following someone through a secure door without authenticating is tailgating (piggybacking).
5	D	With first-match processing, rule 1 allows everything, so the Telnet deny rule is never reached.
6	B	Segmentation limits how far an attacker can move from one compromised device.
7	C	Full-disk encryption makes data unreadable without the key if the device is lost or stolen.
8	A	Hashing is one-way, so a stolen database doesn't directly reveal passwords (especially with salting).
9	D	Users should have only the access their job requires: least privilege.
10	A	Many rapid failed attempts from one source are a classic indicator of a brute-force attack.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Learn the CIA triad, authentication and common attacks, then practice reading logs.

10 KNOW THE TARGET

How students score

AP Cybersecurity is a newer AP course, so national score history is still building. It is scored on the same 1–5 scale as every AP exam.

New

course: national score history is still building

1–5

AP scores use the same five-point scale as every AP exam

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, with the charger packed
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ Plan your time: about **80 s** per multiple-choice question and **50 min** for the analysis question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Review key **controls and attack types** the night before
- ☐ Practice reading a log and a firewall rule set under timing

In the free-response question: read every source, cite specific log lines or rules as evidence, identify asset, threat and vulnerability, use the CIA triad for impact, and explain how each recommended control reduces the risk.



ACT SAT GO • 1-ON-1 AP TUTORING

Aiming for a 5? Get a personal AP Cybersecurity plan.

Book a free diagnostic lesson. We'll find your weakest units, review your practice responses, and give you a week-by-week plan to exam day in May.

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Cybersecurity tutoring

Live 1-on-1 sessions on all five units, with log and firewall analysis drills.

Free-response coaching

Practice evidence-based analysis questions with rubric-style feedback.

Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

◆ Contact us

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